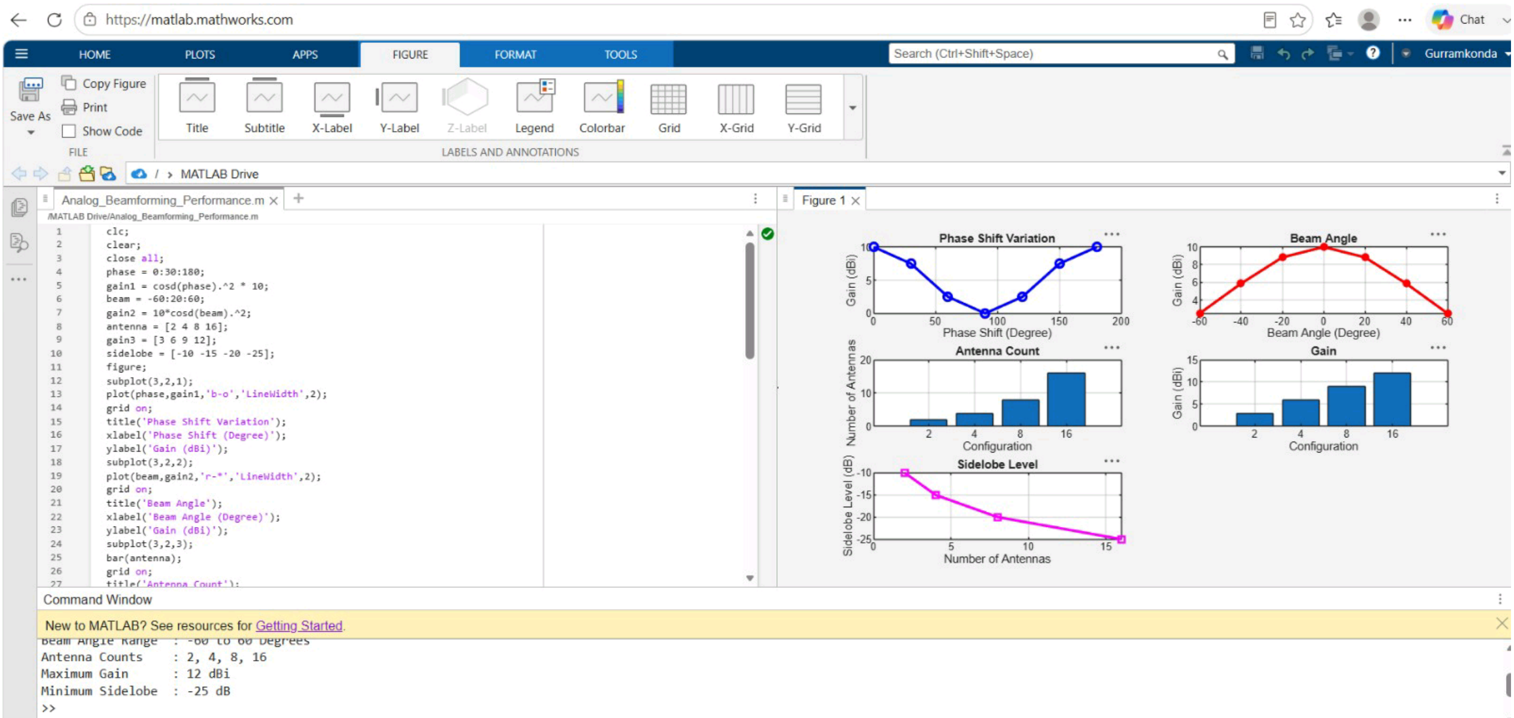
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HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Gurramkonda
Save As
Copy Figure
Print
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Analog_Beamforming_Performance.m
/MATLAB Drive/Analog_Beamforming_Performance.m
clc;
clear;
close all;
phase = 0:30:180;
gain1 = cosd(phase).^2 * 10;
beam = -60:20:60;
gain2 = 10*cosd(beam).^2;
antenna = [2 4 8 16];
gain3 = [3 6 9 12];
sidelobe = [-10 -15 -20 -25];
figure;
subplot(3,2,1);
plot(phase,gain1,'b-o','LineWidth',2);
grid on;
title('Phase Shift Variation');
xlabel('Phase Shift (Degree)');
ylabel('Gain (dBi)');
subplot(3,2,2);
plot(beam,gain2,'r-*','LineWidth',2);
grid on;
title('Beam Angle');
xlabel('Beam Angle (Degree)');
ylabel('Gain (dBi)');
subplot(3,2,3);
bar(antenna);
grid on;
Figure 1
Phase Shift Variation
Gain (dBi)
Phase Shift (Degree)
Beam Angle
Gain (dBi)
Beam Angle (Degree)
Antenna Count
Number of Antennas
Configuration
Gain
Gain (dBi)
Configuration
Sidelobe Level
Sidelobe Level (dB)
Number of Antennas
Command Window
New to MATLAB? See resources for Getting Started.
Antenna Counts   : 2, 4, 8, 16
Maximum Gain   : 12 dBi
Minimum Sidelobe   : -25 dB
>>